

2 Classical (Fixed-Effects) Estimation Theory

2.1 Linear Models

Let $\underline{y}$ represent an $n \times 1$ vector of (scalar-valued) data points. We suppose that

$$\underline{y} = X\underline{\beta} + \underline{\epsilon},$$

where X is a given $n \times p$ “design” matrix, $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters that is contained in some given parameter space, and $\underline{\epsilon}$ is an $n \times 1$ vector of unobservable “residuals” or “deviations.”

It is convenient to introduce various special cases of this “linear model.”

Model I (Least squares model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; parameter space = $\{\underline{\beta}: \underline{\beta} \in \mathbb{R}^p\}$ (where $\mathbb{R}^p$ denotes the Euclidean p -space).

Model II (Gauss-Markov model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = \sigma^2 I$; parameter space = $\{\underline{\beta}, \sigma^2: \underline{\beta} \in \mathbb{R}^p, \sigma^2 > 0\}$.

Model III (Aitken model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given positive definite or positive semidefinite matrix; parameter space = $\{\underline{\beta}, \sigma^2: \underline{\beta} \in \mathbb{R}^p, \sigma^2 > 0\}$.

Model IV (Mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$;

$$\underline{\epsilon} = Z_1 \underline{b}_1 + \cdots + Z_m \underline{b}_m + \underline{e},$$

where Z_i is a given $n \times q_i$ “design” matrix, $\underline{b}_i$ is a $q_i \times 1$ random vector such that $E(\underline{b}_i) = \underline{0}$ and $\text{Var}(\underline{b}_i) = \sigma_i^2 I$, and $\underline{e}$ is an $n \times 1$ random vector such that $E(\underline{e}) = \underline{0}$ and $\text{Var}(\underline{e}) = \sigma^2 I$, and where $E(\underline{b}_i \underline{e}') = 0$ and $E(\underline{b}_i \underline{b}_j') = 0$ for $i = 1, \dots, m$, $j = 1, \dots, m$, and $i \neq j$; parameter space = $\{\underline{\beta}, \sigma_1^2, \dots, \sigma_m^2, \sigma^2: \underline{\beta} \in \mathbb{R}^p, \sigma_1^2 \geq 0, \dots, \sigma_m^2 \geq 0, \sigma^2 > 0\}$. Here, $\sigma_1^2, \dots, \sigma_m^2$, and σ^2 are called variance components.

Model V (Random-effects model): a special case of the mixed-effects model where $X = \underline{1}$ (a column vector of 1's).

Model VI (General mixed-effects model, with a linear variance structure):

$\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = \theta_1 H_1 + \dots + \theta_m H_m$, where $H_1, \dots, H_m$ are given symmetric matrices; parameter space $= \{\underline{\beta}, \underline{\theta}: \underline{\beta} \in \mathbb{R}^p, \underline{\theta} \in \Omega\}$, where $\underline{\theta} = (\theta_1, \dots, \theta_m)'$ and Ω is the set of $\underline{\theta}$ -vectors for which $\text{Var}(\underline{\epsilon})$ is positive definite or positive semidefinite, or Ω is a given subset of that set.

Model VII (General mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = V$, where the elements of V are functions of some m -dimensional parameter vector $\underline{\theta}$; parameter space $= \{\underline{\beta}, \underline{\theta}: \underline{\beta} \in \mathbb{R}^p, \underline{\theta} \in \Omega\}$, where Ω is the set of $\underline{\theta}$ -vectors for which V is positive definite or positive semidefinite, or Ω is a given subset of that set.

Remarks

- Model II is a special case of Model III.
- Models III and IV (and V) are special cases of Model VI.
- Model VI is a special case of Model VII.
- Models II—VII can all be viewed as special cases of Model I.

In all of Models I—VII, the space of possible values for $\underline{\beta}$ is the Euclidean p -space. Limiting $\underline{\beta}$ to some subset of $\mathbb{R}^p$ gives a constrained version of each model. The kinds of limitations on $\underline{\beta}$ that are most commonly imposed are those determined by linear equality constraints or linear inequality constraints, such as $A\underline{\beta} = \underline{b}$ or $A\underline{\beta} \geq \underline{b}$.

Notation. Let $p^* = \text{rank}(X)$.

We have $p^* \leq p$. We do not require that $p^* = p$. When $p^* = p$, the linear model is customarily referred to as a full-rank linear model.

The problem to be considered is that of making inferences about the parameters of the linear model or about various functions of these parameters. In particular, we will consider making inferences about linear combinations of the elements of $\underline{\beta}$, such as obtaining a point estimate or confidence interval for a linear combination $\underline{\lambda}'\underline{\beta}$, or testing the null hypothesis $\underline{\lambda}'\underline{\beta} = c$, where c is a specified constant. More generally, we may seek an estimator or a confidence region for a vector $\Lambda'\underline{\beta}$, where Λ is a given $p \times r$ matrix (so that $\Lambda'\underline{\beta}$ is an r -dimensional vector of linear functions of $\underline{\beta}$), or a simultaneous test of the null hypothesis $\Lambda'\underline{\beta} = \underline{c}$, where $\underline{c}$ is a specified vector.

A related problem is that of making inferences about $\underline{\lambda}'\underline{\beta} + u$, where u is a random variable. For example, $\underline{\lambda}'\underline{\beta} + u$ may represent the model for a future data point. The problem is simplest when u is assumed to be uncorrelated with the elements of $\underline{\epsilon}$. We may wish to predict (or estimate) $\underline{\lambda}'\underline{\beta} + u$ or to establish a prediction (or confidence) interval for it. A generalization of this problem is that of simultaneous inference for a vector $\Lambda'\underline{\beta} + \underline{u}$, where $\underline{u}$ is a random vector. We may seek a prediction (or confidence) region for $\Lambda'\underline{\beta} + \underline{u}$.

2.2 Estimability

Definition. An estimator $t(\underline{y})$ of a linear function $\underline{\lambda}'\underline{\beta}$ is said to be linear if $t(\underline{y}) \equiv c + \underline{a}'\underline{y}$ for some constant c and some vector $\underline{a}$ of constants.

Definition. Under Models II—VII, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] \equiv \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta} \in \mathbb{R}^p$

Note that, under Models II—VII, $E(c + \underline{a}'\underline{y}) =$

Lemma 2.1. *Under Models II—VII, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}'X$.*

Definition. Under Models II—VII, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Otherwise, it is said to be nonestimable.

Theorem 2.1. *Under Models II—VII, a necessary and sufficient condition for a linear function $\underline{\lambda}'\underline{\beta}$ to be estimable is that its coefficient vector $\underline{\lambda}'$ be expressible in the form $\underline{\lambda}' = \underline{a}'X$, that is, that $\underline{\lambda}'$ belong to $\mathcal{R}(X)$, that is, that $\underline{\lambda}'\underline{\beta}$ be expressible as $\underline{a}'X\underline{\beta}$, that is, that*

$$\text{rank}(X) = \text{rank} \begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix},$$

where $\begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix}$ is the matrix X augmented by the row vector $\underline{\lambda}'$.

Proof. Let $\underline{u}'_i$ be the i th row of X and a_i be the i th element of $\underline{a}$. Then,

$$X = \begin{pmatrix} \underline{u}'_1 \\ \vdots \\ \underline{u}'_n \end{pmatrix}, \quad \underline{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad \text{and} \quad \underline{a}'X = (a_1, \dots, a_n) \begin{pmatrix} \underline{u}'_1 \\ \vdots \\ \underline{u}'_n \end{pmatrix} = a_1\underline{u}'_1 + \dots + a_n\underline{u}'_n \in \mathcal{R}(X).$$

$$\text{Thus, } \underline{\lambda}' = \underline{a}'X \iff \underline{\lambda}' \in \mathcal{R}(X) \iff \text{rank}(X) = \text{rank} \begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix}.$$

Corollary 1. Under Models II—VII, the elements of the vector $X\underline{\beta}$ are estimable.

Corollary 2. Under Models II—VII, if $\underline{\lambda}'_1\underline{\beta}, \dots, \underline{\lambda}'_r\underline{\beta}$ are estimable, then any linear combination of them is estimable.

Corollary 3. If $p^* = p$, then for every $p \times 1$ vector $\underline{\lambda}$, $\underline{\lambda}'\underline{\beta}$ is estimable.

We next add a couple of technical matrix results for later use.

Lemma 2.2. *For any matrices A and B (of appropriate dimensions), $\text{rank}(AB) \leq \text{rank}(A)$ and $\text{rank}(AB) \leq \text{rank}(B)$.*

Lemma 2.3. *For any matrix A , $\text{rank}(A'A) = \text{rank}(A)$.*

Question. How many essentially different estimable functions are there?

To answer the question, we make precise the notion of “essentially different.”

Definition. Linear functions $\underline{\lambda}'_1\beta, \dots, \underline{\lambda}'_r\beta$ are said to be linearly independent if $\underline{\lambda}_1, \dots, \underline{\lambda}_r$ form a linearly independent set of vectors or, equivalently, if the matrix $\Lambda = (\underline{\lambda}_1, \dots, \underline{\lambda}_r)$ has rank r .

Theorem 2.2. *Under Models II—VII, there exists a set of p^* linearly independent functions $\underline{\lambda}'_1\beta, \dots, \underline{\lambda}'_{p^*}\beta$ that are estimable. No collection of estimable linear functions can contain more than p^* linearly independent functions.*

Proof. Let $\underline{u}'_i$ be the i th row of X . There exist p^* linearly independent rows of X , say $\underline{u}'_{k_1}, \dots, \underline{u}'_{k_{p^*}}$. By Corollary 1 to Theorem 2.1, $\underline{u}'_{k_1}\beta, \dots, \underline{u}'_{k_{p^*}}\beta$ are p^* linearly independent estimable functions, proving the first part of the theorem.

2.3 Least Squares Estimation

Definition. Under Models II—VII, a (simple or ordinary) least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\hat{\underline{\beta}}$, where $\hat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes the sum of squares

$$Q(\underline{\beta}) = (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta}).$$

A necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \hat{\underline{\beta}}$ is that

$$\left. \frac{\partial Q(\underline{\beta})}{\partial \underline{\beta}} \right|_{\underline{\beta}=\hat{\underline{\beta}}} = \underline{0}.$$

Using vector calculus (see, for example, Section 10.8 of Graybill's book *Matrices with Applications in Statistics*),

$$\begin{aligned} \frac{\partial Q(\underline{\beta})}{\partial \underline{\beta}} &= \frac{\partial(\underline{y}'\underline{y})}{\partial \underline{\beta}} - \frac{\partial(2\underline{y}'X\underline{\beta})}{\partial \underline{\beta}} + \frac{\partial(\underline{\beta}'X'X\underline{\beta})}{\partial \underline{\beta}} \\ &= -2X'\underline{y} + 2X'X\underline{\beta}. \end{aligned}$$

Thus, a necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \hat{\underline{\beta}}$ is that $\hat{\underline{\beta}}$ satisfy the equations

$$X'X\hat{\underline{\beta}} = X'\underline{y}.$$

These equations are called the normal equations.

Remark. (The material in this remark is optional.) The vector-calculus results used above can be obtained as follows. For $\underline{\beta} = (\beta_1, \dots, \beta_p)'$ and any $p \times 1$ vector of constants $\underline{a} = (a_1, \dots, a_p)'$,

$$\frac{\partial \underline{a}'\underline{\beta}}{\partial \underline{\beta}} = \begin{pmatrix} \frac{\partial(a_1\beta_1 + \dots + a_p\beta_p)}{\partial \beta_1} \\ \vdots \\ \frac{\partial(a_1\beta_1 + \dots + a_p\beta_p)}{\partial \beta_p} \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_p \end{pmatrix} = \underline{a}.$$

Thus,

$$\frac{\partial(\underline{y}'X\underline{\beta})}{\partial \underline{\beta}} = \frac{\partial[(X'\underline{y})'\underline{\beta}]}{\partial \underline{\beta}} = X'\underline{y}.$$

Let $\underline{u}'_i$ be the i th row of X . Then,

$$X\underline{\beta} = \begin{pmatrix} \underline{u}'_1 \\ \vdots \\ \underline{u}'_n \end{pmatrix} \underline{\beta} = \begin{pmatrix} \underline{u}'_1 \underline{\beta} \\ \vdots \\ \underline{u}'_n \underline{\beta} \end{pmatrix}$$

and

$$\underline{\beta}' X' X \underline{\beta} = (\underline{u}'_1 \underline{\beta}, \dots, \underline{u}'_n \underline{\beta}) \begin{pmatrix} \underline{u}'_1 \underline{\beta} \\ \vdots \\ \underline{u}'_n \underline{\beta} \end{pmatrix} = \sum_{i=1}^n (\underline{u}'_i \underline{\beta})^2.$$

Thus,

$$\begin{aligned} \frac{\partial(\underline{\beta}' X' X \underline{\beta})}{\partial \underline{\beta}} &= \frac{\partial}{\partial \underline{\beta}} \left(\sum_{i=1}^n (\underline{u}'_i \underline{\beta})^2 \right) = \sum_{i=1}^n 2(\underline{u}'_i \underline{\beta}) \frac{\partial(\underline{u}'_i \underline{\beta})}{\partial \underline{\beta}} = 2 \sum_{i=1}^n (\underline{u}'_i \underline{\beta}) \underline{u}_i \\ &= 2 \sum_{i=1}^n \underline{u}_i (\underline{u}'_i \underline{\beta}) = 2 \sum_{i=1}^n (\underline{u}_i \underline{u}'_i) \underline{\beta} = 2X' X \underline{\beta}. \end{aligned}$$

Theorem 2.3. *The normal equations are consistent.*

Theorem 2.4. $Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$ if and only if $\widehat{\underline{\beta}}$ is a solution to the normal equations.

Proof. For the “if” part ($\Leftarrow$), let $\widehat{\underline{\beta}}$ represent any solution to the normal equations. Then,

$$\begin{aligned} Q(\underline{\beta}) &= (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta}) = [(\underline{y} - X\widehat{\underline{\beta}}) + (X\widehat{\underline{\beta}} - X\underline{\beta})]'[(\underline{y} - X\widehat{\underline{\beta}}) + (X\widehat{\underline{\beta}} - X\underline{\beta})] \\ &= [(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \underline{\beta})]'[(\underline{y} - X\widehat{\underline{\beta}}) + X(\widehat{\underline{\beta}} - \underline{\beta})] \\ &= (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}}) + [X(\widehat{\underline{\beta}} - \underline{\beta})]'[X(\widehat{\underline{\beta}} - \underline{\beta})] + 2(\widehat{\underline{\beta}} - \underline{\beta})'X'(\underline{y} - X\widehat{\underline{\beta}}). \end{aligned}$$

For the “only if” part ($\Rightarrow$), note that $Q(\underline{\beta}) \geq Q(\widehat{\underline{\beta}})$ for all $\underline{\beta}$ with equality holding only if $X(\widehat{\underline{\beta}} - \underline{\beta}) = \underline{0}$, that is, only if $X\underline{\beta} = X\widehat{\underline{\beta}}$. Thus, if $\widetilde{\underline{\beta}}$ is any other value of $\underline{\beta}$ at which $Q(\underline{\beta})$ attains a minimum, then $X\widetilde{\underline{\beta}} = X\widehat{\underline{\beta}}$.

From Theorem 2.4, $t(\underline{y})$ is a **least squares estimate** of an estimable function $\underline{\lambda}'\underline{\beta}$ if and only if $t(\underline{y}) = \underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.

For an estimable function $\underline{\lambda}'\underline{\beta}$, there remains the question of whether $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations, that is, whether $\underline{\lambda}'\underline{\beta}$ has a unique least squares estimate, and, more generally, whether for any function $\underline{\lambda}'\underline{\beta}$, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$. The answer is given by the following theorem.

Theorem 2.5. *Under Models II—VII, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.*

Proof. For the “only if” part ($\implies$), by Theorem 1.3, all solutions to the normal equations are obtained from

$$(X'X)^{-}X'y + [I - (X'X)^{-}X'X]\underline{z}$$

as $\underline{z}$ ranges over the space of all possible vectors. Thus, setting $\widehat{\underline{\beta}} \equiv (X'X)^{-}X'y$ as one solution and $\widetilde{\underline{\beta}} \equiv (X'X)^{-}X'y + [I - (X'X)^{-}X'X]\underline{z}$ as another solution for some $\underline{z}$, if $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ holds for any two solutions of the normal equations, then

$$\underline{\lambda}'[I - (X'X)^{-}X'X]\underline{z} = 0 \text{ for all } \underline{z}.$$

This implies that

$$\underline{\lambda}'[I - (X'X)^{-}X'X] = \underline{0}' \text{ or } \underline{\lambda}' = \underline{\lambda}'(X'X)^{-}X'X = \underline{a}'X \text{ for } \underline{a} = [\underline{\lambda}'(X'X)^{-}X']'.$$

By Theorem 2.1, $\underline{\lambda}'\underline{\beta}$ is estimable.

Corollary. $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations.

2.4 Geometric Considerations

Definition. Two $t \times 1$ vectors $\underline{a}$ and $\underline{b}$ are said to be orthogonal if their inner product $\underline{a}'\underline{b}$ equals zero.

We adopt the following definitions from Graybill's book *Matrices with Applications in Statistics* (Definition 4.2.4).

Definition. Let l_1 be the directed line segment from the origin (zero vector) to a non-zero point $\underline{b}$, and let l_2 be the directed line segment from the origin to a non-zero point $\underline{a}$. The angle θ formed by l_1 and l_2 is defined by the formula

$$\cos \theta = \frac{\underline{b}'\underline{a}}{[\underline{b}'\underline{b}]^{1/2} [\underline{a}'\underline{a}]^{1/2}}, \quad 0 \leq \theta \leq \pi.$$

When $\theta = \pi/2$, that is, when $\cos \theta = 0$, that is, when $\underline{a}$ and $\underline{b}$ are orthogonal, the directed line segments (or the lines defined by the directed line segments) are said to be perpendicular.

Notation. Let $\mathcal{C}^\perp(X)$ represent the null space of X' , that is, the set of all column vectors that are orthogonal to every column of X [or, equivalently, to every vector in $\mathcal{C}(X)$]. Similarly, $\mathcal{R}^\perp(X)$ denotes the set of all row vectors that are orthogonal to every row of X .

Remark. Note that

$$\mathcal{C}^\perp(X) = \{\underline{a} : \underline{a} \in \mathbb{R}^n, \underline{a}'X = \underline{0}'\} = \{\underline{a} : \underline{a} \in \mathbb{R}^n, X'\underline{a} = \underline{0}\}$$

and

$$\mathcal{R}^\perp(X) = \{\underline{b}' : \underline{b} \in \mathbb{R}^p, X\underline{b} = \underline{0}\}.$$

Let $\underline{\hat{y}} = X\underline{\hat{\beta}}$ and $\underline{\hat{\epsilon}} = \underline{y} - \underline{\hat{y}}$, where $\underline{\hat{\beta}}$ is any solution to the normal equations.

- elements of $\underline{\hat{y}}$ are the least squares estimates of corresponding elements of $X\underline{\beta}$.
- elements of $\underline{\hat{\epsilon}}$ can be regarded as estimates of corresponding elements of $\underline{\epsilon}$.
- the data vector $\underline{y}$ admits the decomposition as the sum of $\underline{\hat{y}}$ and $\underline{\hat{\epsilon}}$

$$\underline{y} = \underline{\hat{y}} + \underline{\hat{\epsilon}}.$$

- further, the two components $\underline{\hat{y}}$ and $\underline{\hat{\epsilon}}$ of this decomposition are orthogonal:

$$\underline{\hat{y}}'\underline{\hat{\epsilon}} = (X\underline{\hat{\beta}})'(\underline{y} - X\underline{\hat{\beta}}) = \underline{\hat{\beta}}'X'(\underline{y} - X\underline{\hat{\beta}}) = \underline{\hat{\beta}}'(X'\underline{y} - X'X\underline{\hat{\beta}}) = \underline{\hat{\beta}}'\underline{0} = 0,$$

The next theorem gives a stronger result regarding the decomposition $\underline{y} = \underline{\hat{y}} + \underline{\hat{\epsilon}}$.

Theorem 2.6. *Any $n \times 1$ vector $\underline{y}$ admits a unique decomposition satisfying $\underline{y} = \underline{y}_1 + \underline{y}_2$, where $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^\perp(X)$. Moreover, this decomposition is given by $\underline{y}_1 = \underline{\hat{y}} = X\underline{\hat{\beta}}$ and $\underline{y}_2 = \underline{\hat{\epsilon}} = \underline{y} - \underline{\hat{y}} = \underline{y} - X\underline{\hat{\beta}}$, where $\underline{\hat{\beta}}$ is any solution to the normal equations.*

Proof. As noted above, $\underline{y} = \underline{\hat{y}} + \underline{\hat{\epsilon}}$, and $\underline{\hat{y}} = X\underline{\hat{\beta}} \in \mathcal{C}(X)$. Also,

$$X'\underline{\hat{\epsilon}} = X'(\underline{y} - X\underline{\hat{\beta}}) = X'\underline{y} - X'X\underline{\hat{\beta}} = \underline{0}, \text{ so that } \underline{\hat{\epsilon}} \in \mathcal{C}^\perp(X).$$

It remains to verify the uniqueness of the decomposition.

Theorem 2.6 indicates some properties on the geometry of least squares estimation:

1. it holds that $\underline{y}_1 = \hat{\underline{y}} = X\hat{\underline{\beta}} = P_X\underline{y}$, where P_X is the projection matrix of X .

This means that $\hat{\underline{y}} = X\hat{\underline{\beta}}$ is the “projection” of the vector $\underline{y}$ onto $\mathcal{C}(X)$; mathematically, it’s *the* linear combination of the columns of X that is the closest distance to $\underline{y}$ (i.e., $\|\underline{y} - \underline{t}\|^2 \geq \|\underline{y} - \hat{\underline{y}}\|^2$ for any other $\underline{t} \in \mathcal{C}(X)$.)

2. as another characterization, if some vector $\underline{t} \in \mathcal{C}(X)$ (i.e., is a linear combination of X columns) and satisfies $X'(\underline{y} - \underline{t}) = 0$ (i.e., the difference $\underline{y} - \underline{t} \in \mathcal{C}^\perp(X)$), it must be that $\underline{t} = \hat{\underline{y}} = X\hat{\underline{\beta}} = P_X\underline{y}$.

3. also $\underline{y}_2 = \hat{\underline{\epsilon}} = \underline{y} - \hat{\underline{y}} = \underline{y} - X\hat{\underline{\beta}} = (I - P_X)\underline{y}$, where $I - P_X$ turns out to equal the project matrix (say P_Z) for any matrix Z with columns that span $\mathcal{C}^\perp(X)$

This means that $\underline{y} - \hat{\underline{y}}$ is the “projection” of the vector $\underline{y}$ onto $\mathcal{C}^\perp(X)$; it’s *the* linear combination of the columns of Z that is the closest distance to $\underline{y}$.

4. in plain words, we can decompose a vector $\underline{y}$ into a “best explanation” $\hat{\underline{y}}$ of $\underline{y}$ (i.e., geometrically close) as a linear combination of X columns and a “residual” component $\hat{\underline{\epsilon}} = \underline{y} - \hat{\underline{y}}$ that cannot be linearly captured by X . These two components are orthogonal/perpendicular.

The following example illustrates there is a unique point $\hat{\underline{y}}$ in $\mathcal{C}(X)$ such that every line in $\mathcal{C}(X)$ is perpendicular to the line $\underline{y} - \hat{\underline{y}}$, and this $\hat{\underline{y}}$ is also the unique point in $\mathcal{C}(X)$ that is closest to (has the minimum Euclidean distance from) $\underline{y}$.

Example. Suppose that

$$n = 2, p = 1, \underline{y} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \text{ and } X = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Since $X'X = 2$ and $X'\underline{y} = 5$, the normal “equations” are $2\hat{\beta} = 5$, which has a unique solution $\hat{\beta} = 2.5$. For $\underline{y} = (y_1, y_2)'$ plotted on the (y_1, y_2) -plane, the column space of X is the line that passes through the origin and has a slope of 1 in the plane. Clearly, $\hat{\underline{y}} = X\hat{\beta} = (2.5, 2.5)' \in \mathcal{C}(X)$, and $\hat{\underline{\epsilon}} = \underline{y} - \hat{\underline{y}} = (-0.5, 0.5)' \in \mathcal{C}^\perp(X)$.

Corollary to Theorem 2.6. If W is any matrix with its column space identical to that of the matrix X [that is, $\mathcal{C}(W) = \mathcal{C}(X)$], then $W\hat{\underline{\tau}} = X\hat{\underline{\beta}}$, where $\hat{\underline{\beta}}$ is any solution to $X'X\hat{\underline{\beta}} = X'\underline{y}$ and $\hat{\underline{\tau}}$ is any solution to $W'W\hat{\underline{\tau}} = W'\underline{y}$.